Assignment 3

Problem 1

Perform the following matrix operations in Simulink:

$$\boldsymbol{M_1} = \begin{bmatrix} -3 & -j & 2j \\ 2 & 5 & 0 \\ 1 & 2j & 4j \end{bmatrix};\ \boldsymbol{M_2} = \begin{bmatrix} 2.5 & 1 & j \\ -2 & 3j & 3j \\ 3 & 0 & 4j \end{bmatrix};\ \boldsymbol{v} = \begin{bmatrix} 2 \\ -3j \\ 1 \end{bmatrix}$$

$$M_1 \times v + \sqrt{M_2^T} \times \frac{\sqrt[3]{2.5}}{1.15} \times \sqrt{v} - (M_2 \times M_1^{-1}) \times \sqrt[3]{v} = ?$$

Note: the multiplication is matrix multiplication and not element-wise multiplication

Problem 2

Go back to Lecture 03 on Simulink and complete Example 4. Re-construct the Simulink system. Update the function 'Celsius_To_Fahrenheit_Run.m' so an output message would be displayed in MATLAB Command Window showing the value of the temperature in Fahrenheit.
Hint: you can, but necessarily must, use the command ***disp*** *from MATLAB.* An example would be:

```
Command Window
>> Celsius_To_Fahrenheit_Run
Enter Temperature in Degree Celsius 10
The Temperature value in Fahrenheit is:
    50

fx >>
```

Problem 3

a. Construct a system in Simulink to display a student's letter grade based on their earned points. Student grades are defined according to the following scales:
   F → 0…65
   D → 66…70
   C → 71…81
   B → 82…87
   A → 88…100

Test your system using different examples.

b. Let the user input the grade from MATLAB Command Window and sees the output (the letter grade) also in the Command Window

Problem 4

Consider the signals below:
$x(t)$ has an amplitude of 1 and frequency of 5 rad/s
$y(t)$ has an amplitude of 0.5 and a frequency of 1 rad/s
$z(t)$ has an amplitude of 2 and a frequency of 15 rad/s

a. Perform the following operation and show the result on the scope:

$$4\big(x(t) - y(t)\big) + \sqrt{3} \times (z(t) - 8) - (3x(t) - 2)$$

b. Search for the *Convolution* block diagram and use the one under *Digital Signal Processing (DSP)* library. Perform the following operation and show the result on the scope:

$$4x(t) * y(t)$$

c. Search for the *Step* function is the *Sources* library. Perform the following operation and show the result on the scope: (keep the *step* function settings as default)

$$u(t) * z(t)$$

d. Search for the *ramp* function in *Sources* library. Set the simulation time to 100 instead of the default 10. Perform the following operation and show the result on the scope: (keep the *ramp* function settings as default)

$$ramp(t) * y(t)$$